EEEE4064 Advanced Control System Design

Coursework 1 Guidelines

This coursework deals with the modeling and analysis of a wireless thermopneumatic micromixer* that is illustrated in Figure 1. The thermopneumatic micromixer consists of two inductor-capacitor (LC) wireless heaters, two air-heating chambers, two microdiffuser elements, two loading reservoirs, a zigzag micromixing element, and an outlet hole. The wireless heaters have identical planar spiral inductors, while each capacitor has different dimensions, causing each heater to have a different resonant frequency (f_r). The device is operated by activating the two LC resonant circuits, which behave as frequency-sensitive wireless heaters. Each LC circuit generates heat when exposed to a radiofrequency (RF) magnetic field with a frequency that matches the f_r of the circuit. Therefore, each heater can be selectively activated by tuning the frequency of the external transmitter coil to the corresponding f_r. When a heater is activated, its temperature rises, which causes the temperature of the air-heating chamber to rise as well. This rise in temperature causes the enclosed air to expand, pushing the liquid outside of the reservoir. The air expansion also causes the top surface of the chamber to deform upwards.

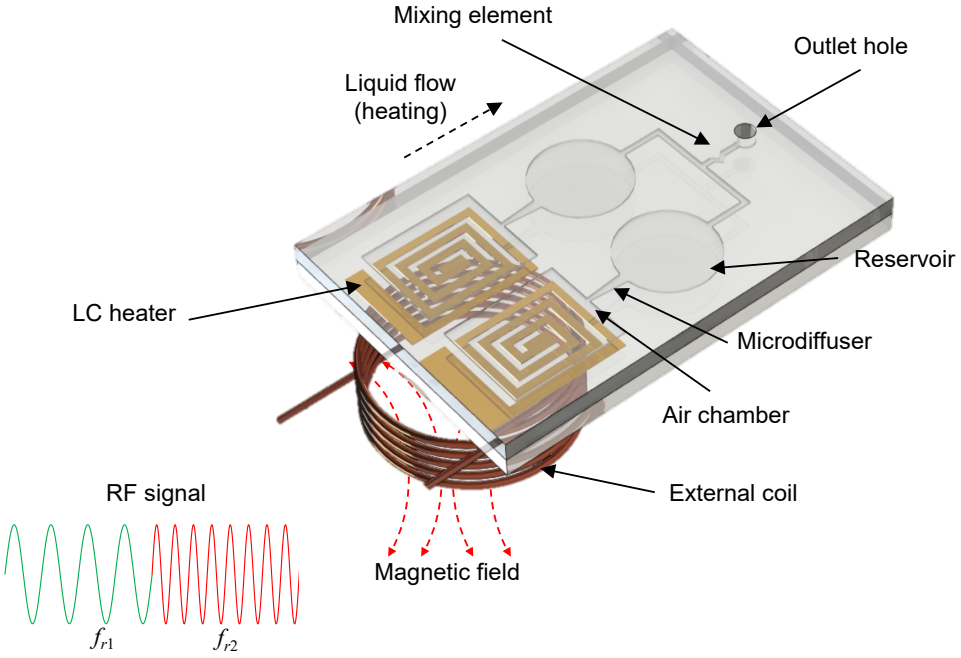


Figure 1. Schematic diagram of the micromixer.

This coursework focuses on a single LC heater and its corresponding air-heating chamber since the only difference between the two pairs is the f_r of the LC heater. Thus, you are required to model and analyze the system despite

* This design appeared in: M. Nafea, A. AbuZaiter, S. Kazi, and M. S. Mohamed Ali, "Frequency-Controlled Wireless Passive Thermopneumatic Micromixer," *Journal of Microelectromechanical Systems*, vol. 26, no. 3, pp. 691–703, 2017. doi.org/10.1109/JMEMS.2017.2692251

not knowing its parameters. Instead, you are provided with experimental data and a Figure that shows its performance (see Figure 2). The figure presents the experimental data achieved by activating one of the LC heaters using an RF signal with a power of 60 mW for 50 seconds, followed by turning it off for the rest of the experiment. The results show the temperature of the LC heater and the resultant deformation of the top surface of the air-heating chamber.

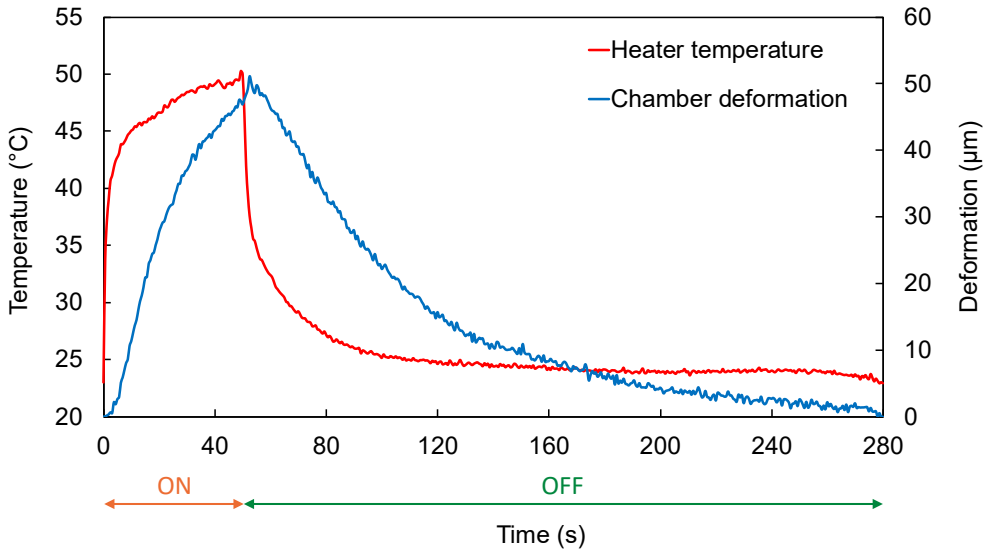


Figure 2. Experimental results of the temperature of the LC heater and the resultant deformation of its corresponding air-heating chamber.

Based on the aforementioned information, you are required to analyze the system after finding its model. You are also required to use the concepts that you have learned in Lectures 1 – 6 and Tutorials 1 – 6. To do that, you need to take the following steps:

1. Download the experimental data file on Moodle and understand the nature of the response of the system.
2. Import the data into MATLAB.
3. Use the System Identification Toolbox in MATLAB to model the system. The lowest acceptable estimation accuracy is 85%.
4. Use the concepts that you have learned in the first 6 weeks of study to model and analyze the system. Use as many concepts as you can to show your deep understanding of the module. This should be done using the formulas presented in the teaching materials.
5. Compare your results from Point 4 with the results found using MATLAB commands for control systems. The results should agree within numerical precision.
6. Analyze the transient response of the identified system model in terms of rise time, settling time, steady-state error, overshoot, and other relevant characteristics.
7. Use comments to present your ideas and to distinguish the different parts of the code. Keep your comments as short as possible. Do not write an essay.

Submission Guidelines

1. Include your name and student ID as a comment at the beginning of your MATLAB code.
2. Save your code as a MATLAB Script file (.m).
3. Save the System Identification session file (.sid).
4. Save the data as a MATLAB data file (.mat).
5. Compress the three files into a single compressed file (.zip).
6. Your MATLAB code must be executable independently using only the provided experimental data and the submitted MATLAB data file.
7. Submit the compressed file via the coursework submission link on Moodle by **10 November 2026**.

Tips

- The beginning of the journey requires additional effort.
- Sometimes, what is missing is just as important as what is given.
- Every matrix has a story to tell.
- Less is more.